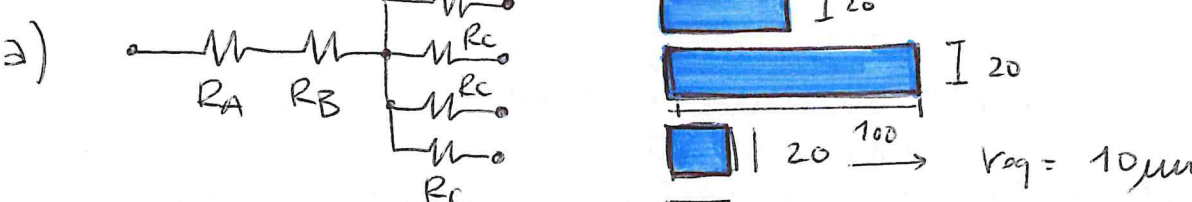


EX 1

7



$$R_A = \frac{12 \cdot \eta \cdot A}{(1 - 0.63 \frac{h}{w}) h^3 w} = \frac{12 \cdot 10^{-3} \cdot \frac{20}{10^{-3}}}{(1 - 0.63 \frac{1}{2}) (20 \mu m)^3 \cdot 40 \mu m} = 5.47 \cdot 10^{13} \left[\frac{Pa \cdot s}{\mu m^3} \right]$$

$$R_B = \frac{12 \cdot \eta \cdot B}{(1 - 0.63 \frac{1}{2}) (20 \mu m)^3 \cdot 40 \mu m} = 1.7 \cdot 10^{13}$$

$$R_C = \frac{8 \eta \cdot C}{\pi r_{eq}^4} = 5 \cdot 10^{14}$$

$$Q = \frac{P_{in} - P_o}{R_{TOT}} = \frac{5 \text{ atm}}{R_A + R_B + R_C/4} = \frac{5 \cdot 101 \text{ kPa}}{19.6 \cdot 10^{13}} = 2.57 \cdot 10^{-9} \frac{m^3}{s}$$

$$Q_A = Q_B = 2.57 \mu l/s \rightarrow V_A = \frac{Q}{h \cdot w_A} = 3.2 \text{ m/s} \quad (2.57 \mu l/s)$$

$$Q_C = \frac{Q_A}{4} = 0.64 \mu l/s \rightarrow V_B = \frac{Q_A}{h \cdot w_B} = 1.2 \text{ m/s}$$

$$\rightarrow V_C = \frac{Q_C}{h \cdot w_C} = 1.6 \text{ m/s}$$

$$t = \frac{10 \mu L}{2.57 \mu l/s} = 3891 \text{ sec} (\sim 1 \text{ hour})$$

1) $T \uparrow \Rightarrow \eta \downarrow \Rightarrow R \downarrow \Rightarrow Q \uparrow$

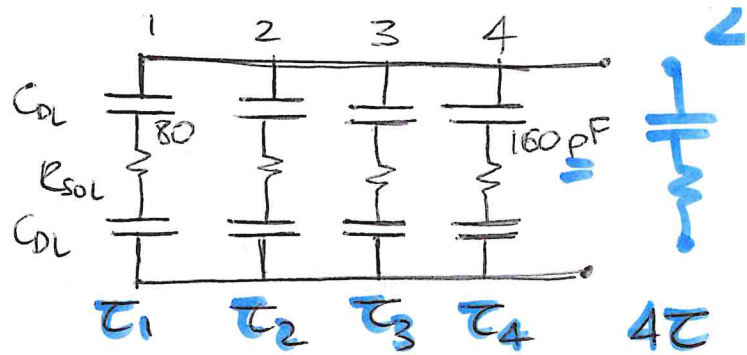
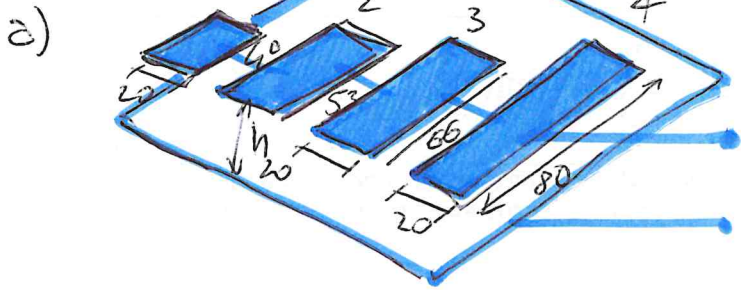
2) DEP $\rightarrow$ NEUTRAL PARTICLES (POLARIZABLE)

EP $\rightarrow$ CHARGED PARTICLES

3) GLASS $\rightarrow$ LIFT-OFF $\rightarrow$ Electrodes

+
SU-8 PHOTORESIST - spin coating
- mask / exposure / development

EX. 2



$$C_{DL} = 0,1 \frac{\text{pF}}{\mu\text{m}^2} \cdot W \cdot L = 0,1 \frac{\text{pF}}{\mu\text{m}^2} \cdot 20 \mu\text{m} \cdot 40 \mu\text{m} = 80 \text{ pF}$$

$$R_{SOL} = \frac{1}{1,5 \text{ S}/\mu\text{m}} \cdot \frac{h}{W \cdot L} = 16,6 \text{ k}\Omega$$

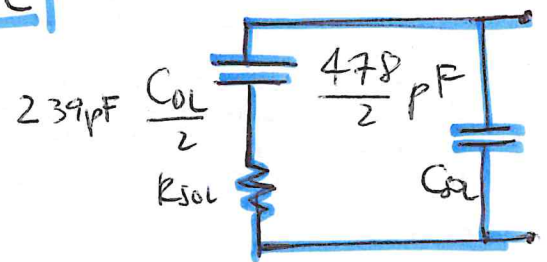
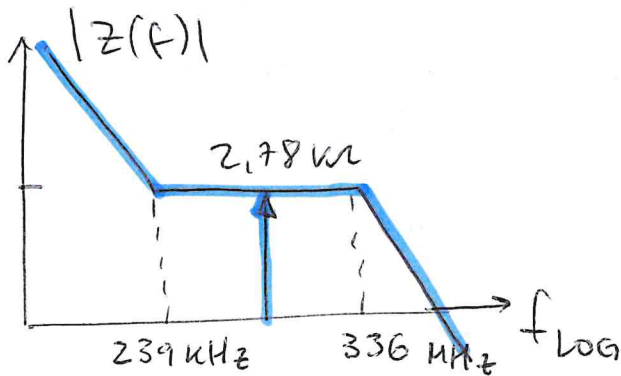
$$\tau = C_{DL} \cdot R_{SOL} \quad \text{INDEPENDENT OF SIZE}$$

$$= \frac{0,1 \text{ pF}}{\mu\text{m}^2} \cdot W \cdot L \cdot \frac{1}{1,5} \cdot \frac{h}{W \cdot L}$$

$$C_{DL} |_{TOT} = \boxed{1 \quad 2 \quad 3 \quad 4}$$

$$= 0,1 \frac{\text{pF}}{\mu\text{m}^2} \cdot W (L_1 + L_2 + L_3 + L_4) = \boxed{478 \text{ pF}}$$

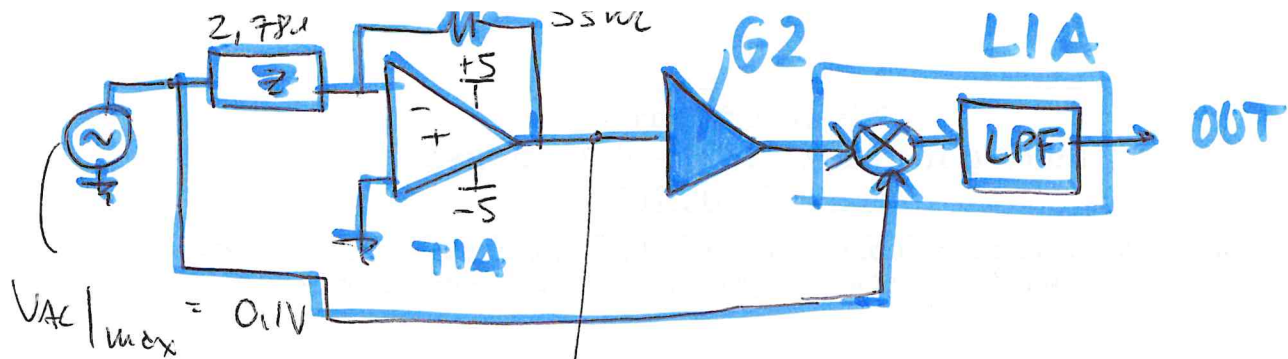
$$R_{SOL} = \frac{1}{1,5 \text{ S}/\mu\text{m}} \cdot \frac{h}{W (L_1 + L_2 + L_3 + L_4)} = \boxed{2,78 \text{ k}\Omega}$$



$$C_{DL} = 8,86 \cdot 10^{-12} \cdot 80 \cdot \frac{W (L_1 + L_2 + L_3 + L_4)}{h} = 0,17 \text{ pF}$$

$$f_{\text{sense}} = 10 \cdot 239 \text{ kHz} = \boxed{2,4 \text{ MHz}}$$

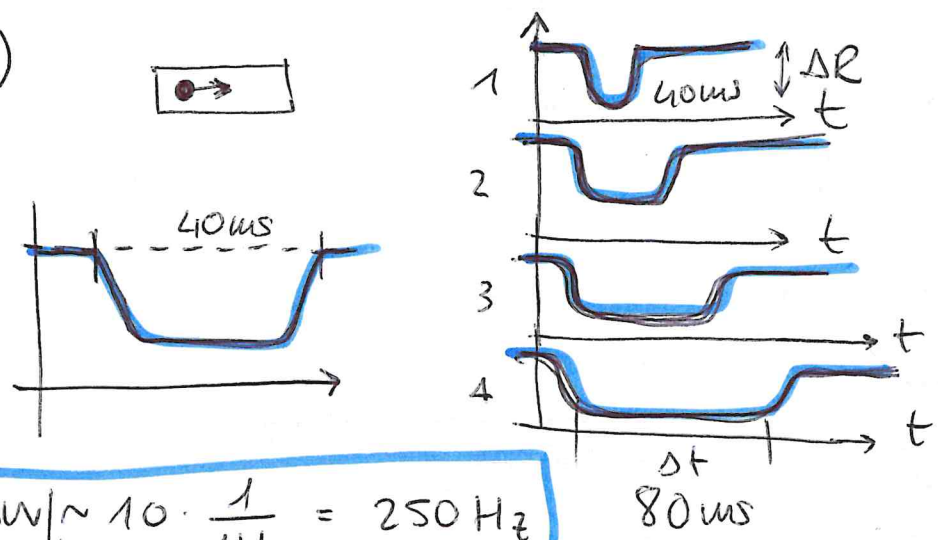
$$f_{\text{BW}} = 10 \cdot f_{\text{SH}} = \boxed{24 \text{ MHz}} = \frac{1}{2\pi C_S R_{\text{depl}}} \Rightarrow R = \frac{1}{2\pi \cdot 0,2 \text{ pF} \cdot 24 \text{ MHz}} = \boxed{33,15 \text{ k}\Omega}$$



$$|V_2| = 0.1V \cdot \frac{33k\Omega}{2.78k\Omega} = 1.2V$$

$$G_2 = \frac{5V}{1.2V} = 4.16 \text{ ADDITIONAL GAIN}$$

c)



$$V = 1 \mu m / ms$$

$$\Delta t = \frac{L}{V} = \frac{40 \mu m}{1 \mu m / ms} = 40 ms$$

$$BW \sim 10 \cdot \frac{1}{\Delta t_{min}} = 250 Hz$$

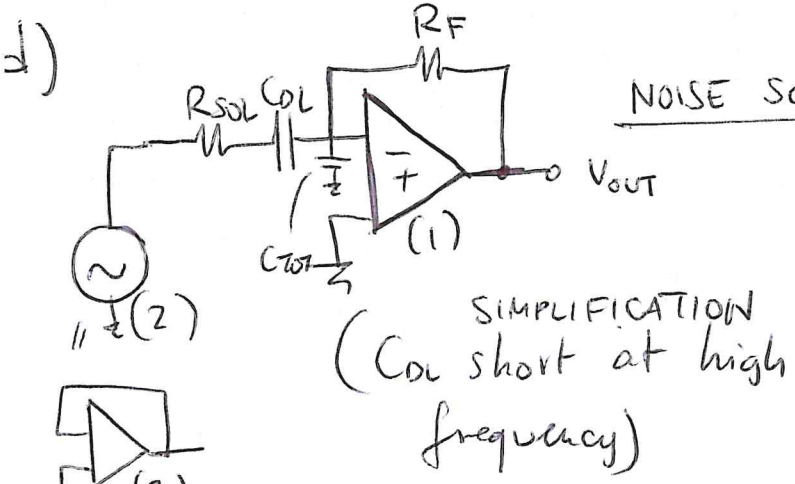
$$\frac{\Delta R}{R} \propto \frac{\Delta Volume}{Vol} \text{ (LINEARIZE)}$$

$$\Delta Volume = \frac{4}{3} \pi r^3 = 5.2 \cdot 10^{-16} m^3$$

$$\frac{\Delta V}{V} = 0.54\%$$

$$Vol = h \cdot w \cdot (L_1 + L_2 + L_3 + L_4) = 9.56 \cdot 10^{-14}$$

$$\Delta R = 2.7k\Omega \cdot 0.54\% \sim 14.6 \Omega$$



NOISE SOURCES

$$R_F \rightarrow \frac{L_{HT}}{R_F}$$

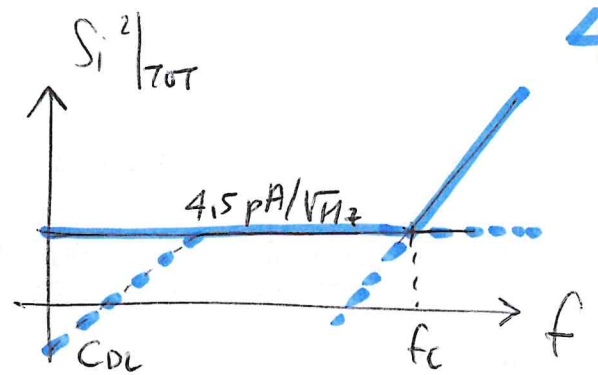
$$e_n |_1 \rightarrow \left(\frac{e_n}{R_F} \right)^2 + (e_n \cdot 2\pi f \cdot C_{TIA})^2$$

$$R_{SOL} \rightarrow \frac{L_{HT}}{R_{SOL}}$$

$$e_n |_2 \text{ VOLTAGE SOURCE } \left(\frac{e_n}{R_{SOL}} \right)^2$$

$$- \frac{4kT}{R_F} = \left(0,7 \frac{\text{pA}}{\sqrt{\text{Hz}}}\right)^2$$

$$- \left(\frac{I_{n1}}{R_{SOL}}\right)^2 = \left(\frac{10 \text{ nV}/\sqrt{\text{Hz}}}{2,7 \text{ k}\Omega}\right)^2 = \left(3,7 \frac{\text{pA}}{\sqrt{\text{Hz}}}\right)^2$$



$$- \frac{4kT}{R_{SOL}} = \left(2,48 \frac{\text{pA}}{\sqrt{\text{Hz}}}\right)^2$$

$$Si^2 |_{TOT} = \frac{4kT}{R_F} + \frac{4kT}{R_{SOL}} + \left(\frac{I_{n1}}{R_{SOL}}\right)^2 = \left(4,5 \frac{\text{pA}}{\sqrt{\text{Hz}}}\right)^2$$

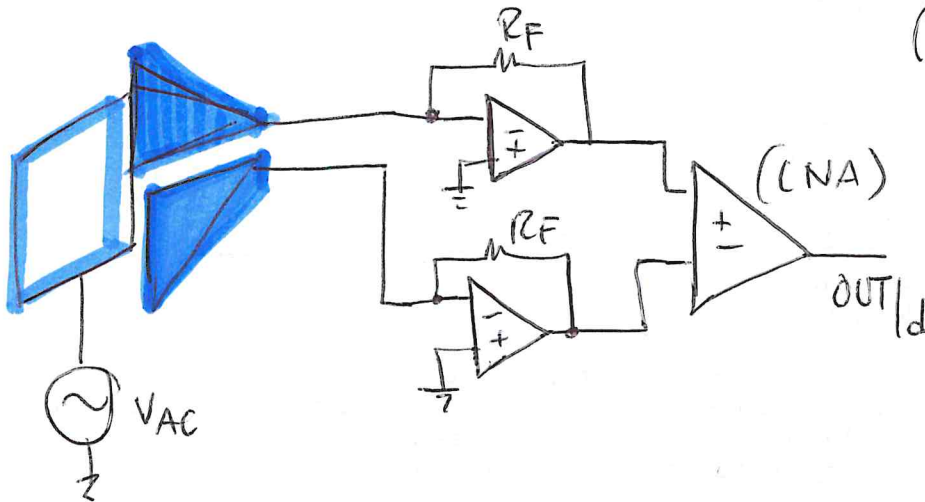
Assuming $C_{TOT} = 1 \text{ pF}$ (0,2 + parasitics)
0,17

$$2\pi f_c C_{TOT} \frac{10 \text{ nV}}{\sqrt{\text{Hz}}} = 4,5 \frac{\text{pA}}{\sqrt{\text{Hz}}}$$

$$f_c = 71,6 \text{ MHz}$$

(above BW/TIA)

d)



ADVANTAGE
↓
REMOVE
SENSITIVITY
TO DRIFT
& VAC NOISE

f)

